

# Continuous Functions

Q1:  $f(x) = \begin{cases} x^2 - 9 & \text{if } x \neq 3 \\ x - 9 & \text{if } x = 3 \end{cases}$   
at  $x = 3$

$$\lim_{x \rightarrow 3} \frac{\cancel{(x-3)}(x+3)}{\cancel{(x-3)}} \Rightarrow (x+3)$$

Applying limit

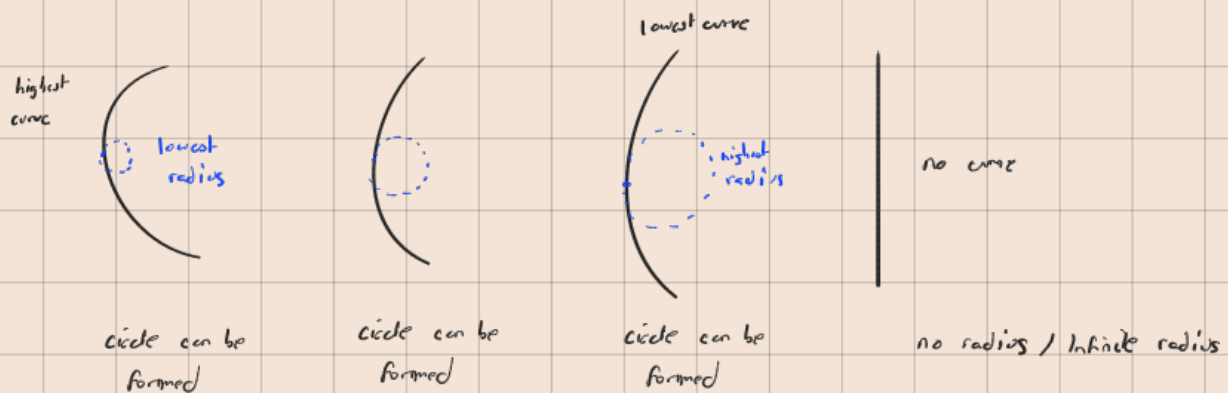
$$(3+3) \Rightarrow \boxed{6} \text{ Ans}$$

Wed Feb 4

## Curvature and Radius of Curvature of a Curve



Curvature is represented by  $\kappa = \text{kappa}$



$$\text{kappa} \propto \frac{1}{\text{radius}}$$

$$\kappa \propto \frac{1}{r}$$

Cartesian form

$(y=x)$

Parametric

$x(t), y(t)$

$$r = \frac{[1 + (y')^2]^{\frac{3}{2}}}{[y'']}$$

$$r = \frac{[(x')^2 + (y')^2]^{\frac{3}{2}}}{|x'y'' - y'x''|}$$

Polar  $r, \theta$

$$r = \frac{[r^2 + (r')^2]^{\frac{3}{2}}}{|r^2 + 2(r')^2 - r r''|}$$

curvature = measure how shortly the curve bends

0)  $y = c \cosh\left(\frac{x}{c}\right)$   $c = \text{constant}$ , h means hyperbolic.

$$y' = \sinh\left(\frac{x}{c}\right) \cdot \frac{1}{c}$$

$$y' = \sinh\left(\frac{x}{c}\right)$$

$$y'' = \cosh\left(\frac{x}{c}\right) \cdot \frac{1}{c}$$
  $\therefore$  no negative sign in hyperbolic

$$y'' = \frac{1}{c} \cosh\left(\frac{x}{c}\right)$$

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$$K = \frac{[1 + (\sinh(\frac{x}{c}))^2]^{\frac{3}{2}}}{|\frac{1}{c} \cosh(\frac{x}{c})|}$$

$$\therefore 1 + \sinh^2 x = \cosh^2 x$$

$$= \left( \cosh^2\left(\frac{x}{c}\right) \right)^{\frac{3}{2}}$$

$$\frac{1}{c} \cosh\left(\frac{x}{c}\right)$$

$$= c \cdot \frac{\cosh^2\left(\frac{x}{c}\right)}{\cosh\left(\frac{x}{c}\right)}$$

$$= c \cdot \cosh\left(\frac{x}{c}\right)$$

$$= c \cdot \cosh^2\left(\frac{x}{c}\right)$$

$$= \frac{c^2 \cdot \cosh^2\left(\frac{x}{c}\right)}{c}$$

$$= \frac{y^2}{c}$$

Qno 2)

$$x = a (\cos t + t \sin t)$$

$$y = a (\sin t - t \cos t) \quad a > 0$$

$$x = a (\cos t + t \sin t)$$

$$x' = a (-\sin t + t \cos t + \sin t)$$

$$x' = a (t \cos t)$$

$$x'' = a (t (-\sin t) + \cos t)$$

$$x'' = (-t \sin t + \cos t)$$

$$y = a (\sin t - t \cos t)$$

$$y' = a (\cos t - (t (-\sin t) + \cos t))$$

$$y' = a (\cos t + t \sin t - \cos t)$$

$$y' = a (t \sin t)$$

$$y'' = a (t \cos t + \sin t)$$

$$r = \frac{[(x')^2 + (y')^2]^{3/2}}{|x'y'' - y'x''|}$$

$$= \frac{[(a(t \cos t))^2 + (a(t \sin t))^2]^{3/2}}{a \cdot t \cos t (a(t \cos t + \sin t)) - a \cdot t \sin t (a(-t \sin t + \cos t))}$$

$$= \frac{[a^2 t^2 (\cos^2 t + \sin^2 t)]^{3/2}}{a^2 t \cos t (t \cos t + \sin t) - a^2 t \sin t (-t \sin t + \cos t)}$$

$$= \frac{[a^2 t^2]^{3/2}}{a^2 t^3}$$

Solving the numerator

$$= [a^2 t^2 \cos^2 t + a^2 t^2 \sin^2 t]^{3/2}$$

$$= [a^2 t^2 (\cos^2 t + \sin^2 t)]^{3/2}$$

$$= [a^2 t^2]^{3/2}$$

$$= a^3 t^3$$

Denominator

$$= a t \cos t (a t \cos t + a \sin t) - a t \sin t (-a t \sin t - a \cos t)$$

$$= a^2 t^2 \cos^2 t + a^2 t \cancel{\cos t \sin t} + a^2 t^2 \sin^2 t - a^2 t \cancel{\sin t \cos t}$$

$$= a^2 t^2 (\cos^2 t + \sin^2 t)$$

$$= a^2 t^2$$

$$= \frac{a^3 t^3}{a^2 t^2} = at$$

Mon, Feb 9

Q3)  $r = a \theta^{-1}$   
 distance  $\rightarrow$  angle  
 $a$  constant

$$r = \frac{a}{\theta}$$

$$r = a \theta^{-1}$$

$$r' = -a \theta^{-2}$$

$$r' = -a \theta^{-2}$$

$$r'' = +2a \theta^{-3}$$

$$r' = -\frac{a}{\theta^2}$$

$$r'' = \frac{2a}{\theta^3}$$

$$\therefore \frac{[r^2 + (r')^2]^{\frac{3}{2}}}{|r^2 + 2(r')^2 - rr''|}$$

$$= \frac{\left[ \left( \frac{a}{\theta} \right)^2 + \left( -\frac{a}{\theta^2} \right)^2 \right]^{\frac{3}{2}}}{\left| \left( \frac{a}{\theta} \right)^2 + 2 \left( -\frac{a}{\theta^2} \right)^2 - \left( \frac{a}{\theta} \right) \left( \frac{2a}{\theta^3} \right) \right|}$$

$$= \frac{\left[ \frac{a^2}{\theta^2} + \frac{a^2}{\theta^4} \right]^{\frac{3}{2}}}{\frac{a^2}{\theta^2} + \frac{2a^2}{\theta^4} - \frac{2a^2}{\theta^4}}$$

$$= \frac{\left[ \frac{a^2}{\theta^2} + \frac{a^2}{\theta^4} \right]^{\frac{3}{2}}}{\frac{a^2}{\theta^2}}$$

$$= \left[ \frac{a^2 \theta^2 + a^2}{\theta^4} \right]^{\frac{3}{2}}$$

$$= \frac{a^{2+\frac{3}{2}} (\theta^2 + 1)^{\frac{3}{2}}}{\theta^{4+\frac{3}{2}}}$$

$$= \frac{a^{\frac{7}{2}} (\theta^2 + 1)^{\frac{3}{2}}}{\theta^{\frac{11}{2}}} \times \frac{\theta^{\frac{7}{2}}}{a^{\frac{7}{2}}}$$

$$\frac{a (\theta^2 + 1)^{\frac{3}{2}}}{\theta^4}$$

$$\theta^{\circ}) \quad r = 2 \cos 2\theta$$

$$\theta = \frac{\pi}{12}$$

$$r = 2 \cos 2 \left( \frac{\pi}{12} \right)$$

$$= 2 \cos \left( \frac{\pi}{6} \right)$$

$$= \sqrt{3}$$

$$r' = 2 (-\sin 2\theta \cdot 2)$$

$$r'' = -8 \cos 2\theta$$

$$r' = -4 \sin 2\theta$$

$$r'' = -8 \cos 2 \left( \frac{\pi}{6} \right)$$

$$r' = -4 \sin \left( \frac{\pi}{6} \right) = -2$$

$$r'' = -4\sqrt{3}$$

$$r = \sqrt{3}$$

$$\therefore \frac{[r^2 + (r')^2]^{\frac{3}{2}}}{|r^2 + 2(r')^2 - rr''|}$$

$$|r^2 + 2(r')^2 - rr''|$$

$$\frac{[(\sqrt{3})^2 + (-2)^2]^{\frac{3}{2}}}{|(\sqrt{3})^2 + 2(-2)^2 - (-2)(-4\sqrt{3})|}$$

$$= \frac{(3+4)^{\frac{3}{2}}}{|3+8-8\sqrt{3}|}$$

$$= \frac{(7)^{\frac{3}{2}}}{|11-8\sqrt{3}|}$$

$$= \frac{10.5}{2.856} = 3.675$$

$$= \frac{21}{2}$$

$$(3+4)^{\frac{3}{2}} = 10.5$$

$$3+8-8\sqrt{3} = -2.856$$

$$10.5 \div 2.856 = 3.675$$

Q)  $r = a(1 + \cos \theta)$  , Prove that  $\frac{p^2}{r} = \frac{3a}{r}$

$$r' = -a \sin \theta \quad r'' = -a \cos \theta$$

$$\frac{[r^2 + (r')^2]^{\frac{3}{2}}}{}$$

$$|r^2 + 2(r')^2 - r r''|$$

$$= \frac{[a^2(1 + \cos \theta)^2 + (-a \sin \theta)^2]^{\frac{3}{2}}}{}$$

$$|a^2(1 + \cos \theta)^2 + 2(-a \sin \theta)^2 - (a(1 + \cos \theta))(-a \cos \theta)|$$

$$= \frac{a^2(1 + 2\cos \theta + \cos^2 \theta) + a^2 \sin^2 \theta}{}$$

$$a^2(1 + 2\cos \theta + \cos^2 \theta) + 2a^2 \sin^2 \theta + a^2(\cos \theta + \cos^2 \theta)$$

$$= \frac{[a^2(1 + 2\cos \theta + \cos^2 \theta + \sin^2 \theta)]^{\frac{3}{2}}}{}$$

$$a^2(1 + 3\cos \theta + 2)$$

$$= \frac{[2a^2(1 + \cos \theta)]^{\frac{3}{2}}}{}$$

$$3a^2(1 + \cos \theta)$$

$$= \frac{2\sqrt{2} a^3 (1 + \cos \theta) \sqrt{1 + \cos \theta}}{3a^2 (1 + \cos \theta)}$$

$$= \frac{2\sqrt{2} \sqrt{1 + \cos \theta}}{3}$$

$$=$$







